

BI Tool & Chart Mapping — Optum Insight, CEO Dashboard

Organization: Optum Insight (UnitedHealth Group segment) **Role / Audience:** CEO, Optum Insight **Source dataset:** `Optum_Insight_CEO_Dashboard_Dataset.xlsx` (star schema — Fact_Revenue, Fact_Cost, Fact_AI_Transactions, Fact_Opportunities, Dim_Date/Product/BusinessLine/Client/State) **Exercise reference:** Day 1 — BI Question-to-Tool Mapping

How each question was worked

For every question the KPI and grain are named first, then the tool, then the chart — never the other way round. Tool choice cites a concrete signal from the tool-landscape table (audience size, refresh cadence, data volume, governance, collaboration). Chart choice is justified against the eazyBI presentation-goal framework (Comparison / Composition / Distribution / Relationship).

Q1 — Revenue & YoY growth by business line

Business question: "Which of our five business lines (RCM/Payer Solutions, Provider Network Solutions, Consulting, Government Solutions, and the newly folded-in Health Financial Services) are driving revenue growth this quarter versus the same quarter last year, and where is growth decelerating?"

KPI / Grain: Revenue (\$) and YoY growth (%), grain = Business Line × Quarter (BL1-BL5 × 2024-Q1...2026-Q2).

Tool recommendation: Power BI. This is a *governed, recurring* board-level metric that must tie exactly to audited/reported figures (the README explicitly flags these numbers as "Reconciled" to real UnitedHealth filings) and gets refreshed and redistributed every quarter to the CEO and the executive committee. That combination — moderate but growing data volume (10+ quarters and climbing), a scheduled refresh cadence, a multi-person recurring audience, and a governance requirement to keep the model auditable and access-controlled — is exactly the signal set the tool-landscape table assigns to Power BI over Excel (too manual/error-prone for a number tied to GAAP disclosures) or Tableau (stronger for ad hoc exploration than for a single certified, distributed dataset).

Chart recommendation: Multi-series line chart (one line per business line, quarters on the x-axis), with YoY growth % shown as data labels/tooltips. Presentation goal = **Comparison** — the question is fundamentally "how does each business line's trajectory compare to the others and to its own prior-year quarter," which is a trend-comparison task, not a composition or distribution task. A line chart lets the CEO immediately see BL3 (Consulting) declining while BL1/BL2 grow, and see BL5 appear only from 2026 due to the segment recast.

Q2 — Net revenue retention (NRR) and client wins/losses by segment and region

Business question: "Which client segments (Payer, Health System, Government, Life Sciences) and regions are expanding or at risk of churning, and how are new-logo wins/losses and renewal churn trending by segment?"

KPI / Grain: $NRR (\%) = \frac{FY2025 \text{ revenue}}{FY2024 \text{ revenue}}$ for the same client cohort, grain = Segment or Region $\times$ Fiscal Year; plus new-logo win/loss counts and churned ACV (\$K), grain = Segment $\times$ Fiscal Year (from Fact_Opportunities).

Tool recommendation — genuine two-tool trade-off, final call: Power BI. This question legitimately supports two tools:

- **Tableau's case:** the underlying question invites free-form exploration (drilling from segment into region into individual client), and Tableau's mapping/geo-visualization and drag-and-drop exploratory authoring are best-in-class for a CEO who wants to poke at "why is the Northeast below target" interactively.
- **Power BI's case:** the analysis requires joining three fact/dimension objects (Fact_Revenue, Dim_Client, Dim_State, Fact_Opportunities) into one governed semantic model, and the output needs to sit alongside Q1's certified revenue dataset and refresh on the same governed schedule for the same recurring executive audience.

Final call — Power BI, because the deciding signal here is *governance and single-source-of-truth consistency*: this KPI is consumed by the same recurring, multi-stakeholder audience as the other CEO KPIs and must reconcile to the same certified data model rather than living in a separate ad hoc workbook/environment. Tableau remains the better choice only if this became a standalone, one-off deep-dive for the CS/regional leadership team rather than a recurring CEO KPI — that is the trade-off to flag explicitly.

Chart recommendation: Diverging bar chart, one bar per segment/region, benchmarked at 100% NRR. Presentation goal = **Comparison** (to a fixed benchmark) — every segment/region needs to be ranked above or below the 100%-retention line, so a diverging bar

(green right of 100%, red left of it) reads faster than a table of eight percentages. A choropleth map by state is a reasonable secondary/drill-through view (this is the geographic angle that strengthens Tableau's case above) but the primary CEO-level visual is the segment/region comparison, not the geography itself.

Q3 — Product portfolio: growth vs. margin

Business question: "Across our 12 products, which are growing fast but eroding margin, which are mature cash generators, and where should we prioritize investment or triage?"

KPI / Grain: Revenue growth % (FY2024→FY2025), operating margin % (FY2025), and revenue \$ (bubble size), grain = Product × Fiscal Year (12 products, P01-P12; P13/P14 excluded pre-2026).

Tool recommendation: Tableau. This is the one question in the set that is genuinely small in data volume (12 rows) and *exploratory/strategic* rather than operational — a periodic portfolio review for the CEO and a handful of strategy leads, not a broadly distributed recurring report. The tool-landscape signal that tips this to Tableau over Power BI is audience size (small, single-digit strategic audience) combined with the need for rich, immediately interactive visual exploration (hovering/filtering bubbles by product family) rather than governed mass distribution — Tableau is purpose-built for this kind of "spot the pattern" analytical chart.

Chart recommendation: Bubble (scatter) chart — x = revenue growth %, y = operating margin %, bubble size = FY2025 revenue, one bubble per product. Presentation goal = **Relationship** — the question is explicitly about the relationship between two continuous variables (growth and margin) for each product, with a third variable (revenue) encoded as size. This is the textbook Relationship use case from the framework, and it matches the dataset's own KPI_Preview tab, which stages this exact data as "bubble chart data: x = growth, y = margin, size = revenue."

Q4 — Operating margin & cost-to-serve vs. FY2026 guidance

Business question: "What is our operating margin trajectory by business line, and are we on track to hit the FY2026 adjusted-margin guidance of 22.6%?"

KPI / Grain: Operating margin % (recurring, excluding one-time items), grain = Business Line × Quarter, benchmarked against a single FY2026 target line (22.6%, real company guidance).

Tool recommendation: Power BI. Every signal here points the same direction: this KPI is tied to **public guidance** (so accuracy/auditability matters as much as Q1), it is **broadly distributed** to the board, CFO, and business-line leaders (not just the CEO alone), it **refreshes every quarter close** on a fixed schedule, and it draws from **two joined fact tables** (Fact_Revenue and Fact_Cost, with one-time items separated out). That mix of governance + broad recurring audience + a standardized refresh cadence is precisely the Power BI zone on the tool-landscape table, as opposed to Excel (too manual for board-distributed, audit-sensitive numbers) or Tableau (better suited to exploratory, smaller-audience analysis as in Q3).

Chart recommendation: Combo/line chart — one line per business line trending operating margin % by quarter, plus a flat reference line at the 22.6% FY2026 target. Presentation goal = **Comparison** — specifically comparison-to-benchmark over time. The CEO needs to see at a glance which business lines are tracking toward guidance and which (e.g., BL4, which dips near 1% in 2026-Q1) are dragging the blended margin down.

Q5 — AI automation rate & client savings

Business question: "How is AI/automation adoption progressing across our AI-enabled products (Autonomous Coding & CDI, Payment Integrity, Digital Prior Authorization), and are we on pace to hit the 65% automation-rate target while growing client savings?"

KPI / Grain: Automation rate (%) and client savings (\$mm), grain = AI Product × Month (Fact_AI_Transactions), rolled up to an "all AI" blended rate and benchmarked against a 65% target.

BI vs. OLTP/OLAP distinction (explicitly applied here): Fact_AI_Transactions is documented in the dataset as "aggregated from transaction-level OLTP data in a real build" — meaning the true source system is an operational, row-per-transaction OLTP database (every individual coding/claims transaction the AI engine processes), which is optimized for fast writes and single-record lookups, not for trend analysis. A BI tool should never query that live OLTP transaction log directly for a CEO KPI; instead it should sit on top of a pre-aggregated, OLAP-style monthly fact table (client × product × month), exactly as this dataset is structured. This separation — OLTP for operational transaction processing, OLAP/BI-ready aggregates for executive reporting — is what keeps a CEO dashboard fast and stable regardless of live transaction volume, and it's the reason this fact table is already shipped as a monthly rollup rather than raw transaction rows.

Tool recommendation: Power BI. Same governance/recurring-audience logic as Q1 and Q4 — this is a standing CEO KPI reported on a fixed cadence, pulling from a governed aggregate (OLAP-style) fact table rather than raw transactional data, and needs to sit in the same

certified semantic model as the other four KPIs for one consistent CEO dashboard rather than a separate one-off file.

Chart recommendation: Dual-axis combo chart — line for blended automation rate % vs. the 65% target line (left axis), stacked columns for client savings \$mm by product (right axis), both by quarter. Presentation goal = **Comparison + Composition together** — the rate-vs-target line answers a Comparison question (are we ahead of or behind the 65% goal), while the stacked columns answer a Composition question (which product — P02, P05, or P06 — is contributing most of the client savings this quarter). A dual-axis combo is a single, legitimate chart type in both Power BI and Tableau that lets both goals sit on one visual without forcing two separate charts onto the CEO's one-page view.

Summary table

#	Business question	KPI / Grain	Tool	Chart
Q1	Revenue & YoY growth by business line	Revenue \$, YoY % — BL × Quarter	Power BI	Multi-series line chart
Q2	NRR & client wins/losses by segment/region	NRR % — Segment/Region × FY	Power BI (<i>two-tool trade-off vs. Tableau — see above</i>)	Diverging bar vs. 100% benchmark
Q3	Product growth vs. margin	Growth %, Margin %, Revenue \$ — Product × FY	Tableau	Bubble/scatter chart
Q4	Operating margin vs. FY2026 guidance	Margin % — BL × Quarter vs. target	Power BI	Line chart vs. benchmark
Q5	AI automation rate & client savings	Automation %, Savings \$mm — Product × Month	Power BI	Dual-axis combo (line + stacked columns)

Prepared for the Day 1 BI Visualization Accelerator exercise — no BI software build required at this stage; this is the tool/chart reasoning that precedes development.